

Research Article

Children's Oral Narrative Skills Through the Lens of Internal State Terms: A Large-Scale Study of Fictional Stories Elicited With the Multilingual Assessment Instrument for Narratives

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ABSTRACT

Purpose: This article investigates the development of oral narrative skills in children through the lens of internal states in a large cross-linguistic data set of comparable fictional stories.

Method: In a joint effort of research groups around the world, 5,507 picture-based stories were elicited from 2,763 monolingual and bilingual typically developing (TD) children aged 3–13 years in 32 countries across 44 languages with the Multilingual Assessment Instrument for Narratives (MAIN; Gagarina et al., 2019). These were transcribed and coded for internal state terms (ISTs) that express story characters' thoughts, emotions, and perceptions. Using generalized mixed models, we investigated the likelihood of a child producing ISTs in the stories, as influenced by age, monolingualism versus bilingualism, and, for bilinguals, first language versus second language, elicitation mode (telling vs. retelling), IST type, story type, and story length.

Results: We found strong positive effects of age and elicitation mode (retelling) on IST production, but no differences between monolinguals and bilinguals or for bilinguals in their two languages. Children were far more likely to produce ISTs as initiating events than ISTs as reactions. Story length only predicted IST production in younger children, but not in older ones.

Conclusions: As shown by this large-scale sample, children's skills in making explicit the internal states of characters in fictional stories develop substantially during the primary school years, across languages, and for both monolingual and bilingual learners. The robust facilitative effect of retelling suggests that retelling may be particularly suitable to use with MAIN when researchers/clinicians are interested in assessing a child's ability to verbalize internal states. Future research should examine ISTs in children with developmental disorders and explore their application in intervention.

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This article investigates children's narrative skills through the lens of internal states in a large sample of comparable fictional stories told or retold by children around the world. Internal state terms (ISTs) express internal states, or mental states, such as thoughts, emotions, perceptions, desires, and beliefs, such as *happy*, *angry*, *sad*, *scared*, *glad*, *pleased*, *satisfied*, *keen*, *disappointed*, *worried*, *curious*, *hungry*, *anxious*, *proud*, *relieved*, *ashamed*, *feel*, *think*, *wonder*, *know*, or *want* (Bamberg & Damrad-Frye, 1991; Fusté-Hermann et al., 2006; Grazzani & Ornaghi, 2012; Trabasso et al., 1992). ISTs make explicit reference to the internal state of an animate being. They can either be single words (e.g., *happy*, *hungry*) or multiword expressions (e.g., *beaming with joy*, *wanting to eat*, *thinking "yummy"*). ISTs are indispensable for the telling and understanding of narratives since they connect episodes, move the plot forward, and provide a window into causal relationships. They are a crucial means for a storyteller to convey implicit story content and show emotional engagement (Bruner, 1986). ISTs also help the listener to understand what language skills a storyteller has to

infer and verbalize the internal states "hidden" in the actions of characters in (fictional) narratives.

Much research has been done on children's understanding of internal states within social cognition, developmental psychology, and related fields (e.g., Antonietti et al., 2006; Astington & Pelletier, 2005; Bartsch & Wellman, 1995; Bretherton & Beeghly, 1982; Grazzani & Ornaghi, 2012; Wimmer & Perner, 1983). However, in the present article, we are concerned with children's narrative skills through the lens of ISTs. ISTs are of interest to researchers studying children's narrative development due to their multifaceted functions, which include the following. First, ISTs are often narrative catalysts: They frequently initiate events or motivate plot progression by indicating a character's intentions or how they express reactions (Berman & Slobin, 1994; Westby, 2012). Second, ISTs reflect children's inferencing abilities, as their appropriate use often requires interpreting nonexplicit or visually implied information, such as the mental or emotional states of characters depicted in

illustrations (Bohnacker & Gagarina, 2020; Pelletier & Astington, 2004). ISTs may also help differentiate between TD children and those with developmental language disorder (DLD), who have been shown to produce fewer and less diverse ISTs, reflecting limitations in inferencing and narrative coherence (Blom & Boerma, 2016; Boerma et al., 2016; Tsimpli et al., 2016). Finally, ISTs shape the expressive and affective quality of narratives, contributing to their dramatic depth and listener engagement (Bamberg & Damrad-Frye, 1991; Newman & McGregor, 2006). Given these multifaceted roles, ISTs offer a valuable lens through which to examine both linguistic and cognitive dimensions of narrative development across diverse populations and contexts. While ISTs feature in different types of narrative discourse, the use of picture-based fictional narratives offers an additional advantage of controlling the elicitation of ISTs via the visual stimuli offered to a child.

Cross-linguistic work on ISTs in fictional narratives is rare, as are bilingual studies, and large-scale investigations are completely missing. This is a knowledge gap that the present study seeks to fill. We systematically investigate the acquisition of ISTs in a very large sample of oral fictional narratives, elicited in a comparable manner from preschool and primary school children from 32 countries.

Early Explorations of the Use of ISTs

Interest in the use of ISTs in fictional narratives emerged decades ago. Much of the earlier narrative research found that children develop ISTs between the ages of 3 and 9 years but also that TD 9-year-olds do not yet use ISTs as often and varied as adults, even though their fictional narratives are well organized (Bamberg & Damrad-Frye, 1991; Berman & Slobin, 1994; Pearson, 2002; Stein, 1988). Seminal studies by Berman and Slobin (1994) and colleagues investigated English, German, Hebrew, Spanish, and Turkish using the same wordless picture book *Frog, Where Are You?* (Mayer, 1969). Other studies of children's ISTs in fictional narratives looked predominantly into English using other materials (e.g., Hudson & Shapiro, 1991; Paris & Paris, 2003; Pelletier & Astington, 2004; Stein, 1988; Stein & Glenn, 1979). In general, there appeared to be a discrepancy between children's production of ISTs in fictional narratives and theory of mind (ToM; i.e., the ability to understand others' thoughts and feelings) as shown in experimental tasks: Children aged 4–5 years were able to solve ToM false-belief tasks and were able to successfully predict some other person's behavior by imagining their thoughts, beliefs, and desires—that is, their internal states (Bartsch & Wellman, 1995). By contrast, children of the same age hardly ever explicitly mentioned the internal states of story characters in narratives. However, as Nicolopoulou and Richner (2007) point out, this might be an artifact of the narrative elicitation

methods used in previous studies, which may not capture the ability to produce ISTs. This is because such elicitation methods may provide children with limited opportunities to express ISTs, which can lead to a less accurate assessment of their actual narrative abilities.

Indeed, previous studies have largely used narrative elicitation materials that are convenience based and not linguistically motivated, such as assorted wordless picture books, for example, *Frog, Where Are You?* and sequel books (Mayer, 1967, 1969, 1974). Alternatively, culture-specific single pictures, cartoons, or other picture sequences have been used that often cannot straightforwardly be used in other cultures and regions of the world, for example, the *Bus Story Test* (Renfrew, 1969/1997) about a typical U.K. double-decker bus or the *Test of Narrative Language* (Gillam & Pearson, 2004) about eating at an American fast-food chain and an alien spaceship landing in a park. Furthermore, such elicitation materials do not include sufficiently parallel pictorial stimuli to assess the two languages of bilingual children in a comparable manner (Fiestas & Peña, 2004). Against this backdrop, a new standardized narrative elicitation and assessment tool, the Multilingual Assessment Instrument for Narratives (MAIN; Gagarina et al., 2012), was developed with parallel picture sequences that are suitable across languages and cultures and that also target ISTs, as described in the next section.

The MAIN and How ISTs Feature in It

The MAIN (Gagarina et al., 2012) was developed by an interdisciplinary and pluricultural research group of linguists, developmental psychologists, and speech-language pathologists from over 20 countries (from Northern Europe to South Africa) who were involved in creating, piloting, revising, and testing MAIN between 2009 and 2013 as part of the COST Action IS0804 *Language Impairment in a Multilingual Society: Linguistic Patterns and the Road to Assessment*. In its revised form (Gagarina et al., 2019), it is now a standardized open-access tool, which is successfully used by more than 3,200 registered users in 100 languages in over 60 countries, primarily with monolingual and bilingual children aged 3–12 years. MAIN includes four parallel picture-based stories (*Cat, Dog, Baby Birds, and Baby Goats*) controlled for protagonists' internal states and also includes scoring sheets, examples of well-formed narratives, instructions for use, and short parental questionnaires (<https://main-leibniz-zas.de>). With the launch of MAIN, it became possible to study the development of fictional narratives not only in a controlled but also in a reliable and comparable way across different languages, language settings, and cultures. By language settings, we mean monolingual and bilingual acquisition, and in the latter, the development of the home language (first language [L1]) and of

the societal language (second language [L2]). The tool is grounded in a theoretical model of narrative macrostructure as consisting of units that are temporally and causally related (Bohnacker & Gagarina, 2020; Gagarina et al., 2015). These units are episodes, which in turn are made up of the core story grammar elements: goal, attempt, outcome, and framing elements—internal states, for example, expressions referring to characters’ emotions, thoughts, desires, and perceptions. For instance, in the first episode of the MAIN *Cat* story (see Figure 1), there is a cat who wants to catch a butterfly (goal); it then jumps forward (attempt) but falls into a thorny bush while the butterfly escapes (outcome). The internal state of the protagonist (the cat’s playfulness, curiosity, awareness, and wish to catch) serves as a trigger for the event chain of the episode. In MAIN, this is called “IST as an initiating event.” Similarly, the second framing element of an episode is the internal state of the protagonists after the event (in this case, the cat’s pain, anger, or disappointment). In MAIN, this is called “IST as a reaction.” As each story consists of three episodes, there are three possibilities to produce ISTs as initiating events and three possibilities to produce ISTs as reactions per story, a maximum of six macrostructurally relevant ISTs, as illustrated in Table 1 for the *Cat* story.

Similarly, for the MAIN *Baby Birds* story, the narrative macrostructure organized into three episodes is illustrated in Figure 2 and Table 2 with a focus on ISTs as initiating events and reactions. For example, in the third

episode of the *Baby Birds*, starting from Picture 4, there is a dog who wants to stop the cat and save the chicks (goal); it then pulls the cat’s tail (attempt) and chases the cat away (outcome). The internal state of the protagonist (the dog who sees that the chicks are in danger) is an initiating event in the last episode. The internal states as reactions and the conclusion for the whole story include the following: the dog being relieved, the cat being scared, the chicks being safe, and the mother bird being happy (for more examples, see Table 2).

ISTs make explicit reference to the internal state of animate beings. As illustrated in Tables 1 and 2, they can be single words or multiword expressions (this may also vary between languages). We have given examples of English ISTs representing the conceptual meaning behind the targeted internal states. These need to be inferred from the pictorial stimuli of MAIN, which consist of four comparable stories, where the *Dog* story (not depicted here) is closely parallel to the *Cat* story, and the *Baby Birds* story and *Baby Goats* story (not depicted here) are closely parallel. The targeted internal states include physiological states (such as being hungry), affective or emotional states (such as being happy, glad, sad, angry, scared, or disappointed), mental states (such as knowing or thinking), and perception states (such as seeing, hearing). There is no expectation for the child to produce the exact lexical expression or an identical sequence of forms as given in Tables 1 and 2 as long as one of the targeted internal states is verbalized. Whereas

Figure 1. Small-scale version of the picture sequence for the Multilingual Assessment Instrument for Narratives *Cat* story (Gagarina et al., 2012).

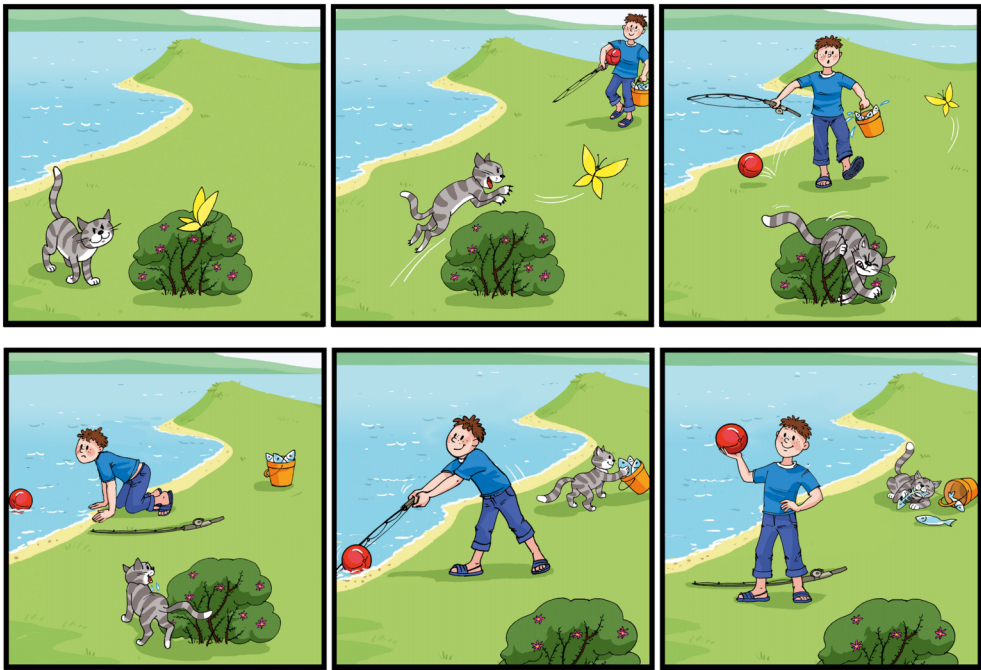


Table 1. Examples of internal state terms (ISTs; in bold) as initiating event and as reaction in the Multilingual Assessment Instrument for Narratives Cat story (scoring sheets in Gagarina et al., 2019, pp. 10–11).

Episode	Cat story	
	IST as initiating event	IST as reaction
Episode 1 characters: cat and butterfly	Cat was playful/ curious Cat saw a butterfly (Picture 1)	Cat was disappointed/ angry/ hurt Butterfly was happy/ glad (Picture 3)
Episode 2 character: boy	Boy was sad/ unhappy/ worried about his ball Boy saw the ball in the water (Picture 4)	Boy was glad/ happy/ pleased/ satisfied/ relieved (to get/have his ball back) (Picture 6)
Episode 3 character: cat	Cat was hungry/ curious/ keen on the fish Cat noticed/ saw the fish (Picture 4)	Cat was satisfied/ glad/ pleased/ happy/ not hungry (any more) (Picture 6)

the exact linguistic form for verbalizing internal states may differ between languages and cultures, the conceptual content in the pictorial stimuli remains constant.

While the picture sequences above (see Figures 1 and 2) show the final version of the stories, a great deal of work lies behind these seemingly simple illustrations. It was a challenge for the MAIN author team to develop the best pictorial representations that would reliably elicit the targeted internal states from children of diverse backgrounds, given that these internal states must be inferred. We give one example: The final scene of the *Baby Birds* story (Episode 3) depicts a dog chasing a cat (see Figure 2). This last picture of Episode 3 is at the same time the end of the whole story. The internal state of the dog is twofold here: On the one hand, it is angry, fierce, and aggressive because it is chasing the cat away from the

chicks. On the other hand, the dog is relieved and happy to be able to save/rescue the chicks. This internal state must also be inferred, although the dog’s face does not show this; *the inference must be drawn from the whole story*. In an earlier version of the illustration (see Figure 3, left image), the dog’s facial expression could be interpreted as joyful or playful, as indicated by an open mouth resembling a smile and relaxed eyebrows. During piloting, some children who otherwise correctly understood the narrative and produced appropriate stories nonetheless interpreted the final scene as the dog playing with the cat instead of chasing the cat away. This misinterpretation highlights the important role of facial expressions in conveying internal states and aligning visual input with narrative intentions. For the final version, the dog’s facial features were revised in the following way: The eyebrows were made thicker and angled downward, the mouth was enlarged, and prominent

Figure 2. Small-scale version of the picture sequence for the Multilingual Assessment Instrument for Narratives Baby Birds story (Gagarina et al., 2012).

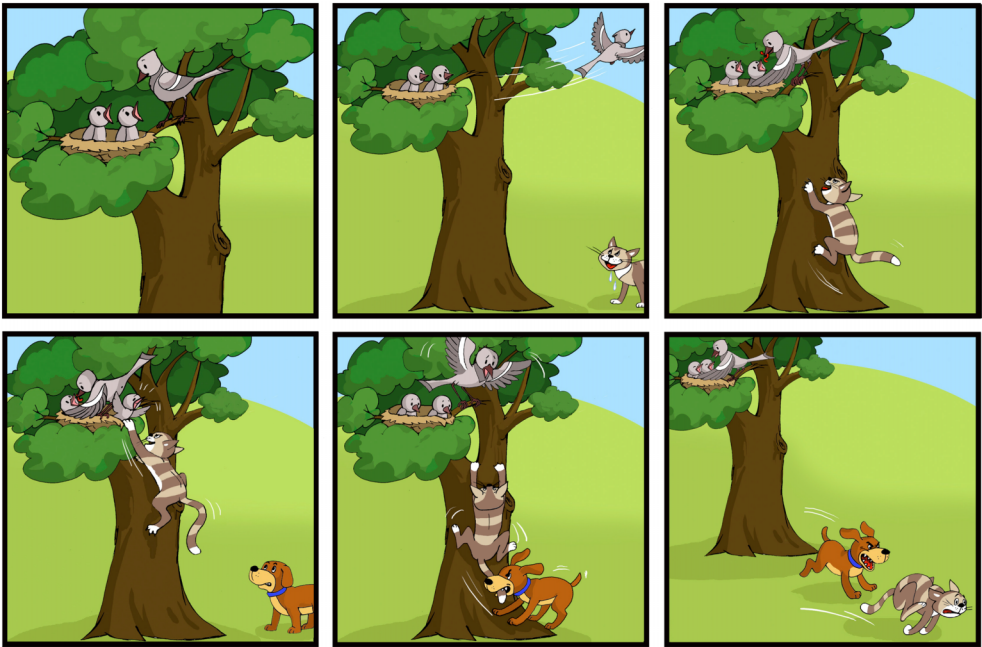


Table 2. Examples of internal state terms (ISTs; in bold) as initiating event and as reaction in the Multilingual Assessment Instrument for Narratives Baby Birds story (scoring sheets in the work of Gagarina et al., 2019, pp. 24–25).

Episode	Baby Birds story	
	IST as initiating event	IST as reaction
Episode 1 characters: mother bird and baby birds	Baby birds were hungry/ wanted food/ cried for food/ asked for food Mother/ Bird/ Parent, etc. saw that baby birds were hungry/ wanted food (Picture 1)	Mother bird was happy/ satisfied/ pleased Baby birds were happy/ satisfied/ pleased/ not hungry anymore (Picture 3)
Episode 2 character: cat and baby bird(s)	Cat saw mother flying away/ saw that baby birds were all alone/ saw that there was food Cat was hungry/ thought “yummy” (Picture 2)	Cat was happy Bird/–s was/ were scared/ crying/ screaming with pain (Picture 4)
Episode 3 characters: dog, cat and baby bird(s)	Dog saw that the bird was in danger/ saw that cat caught/ got the bird Bird/–s was/were in danger (Picture 4)	Dog was relieved/ happy/ proud (to have saved/ rescued the baby bird) Cat was angry/ disappointed/ feeling bad/ mad/ scared/ in pain/ cat's tail hurt Bird/–s was/ were relieved/ happy/ safe Mother bird was relieved/ happy (Picture 6)

sharp teeth were added. This visual enhancement aimed to make the dog’s aggressive intention unmistakable, thereby supporting more accurate narrative interpretation and production (see Figure 3, right image). We would like to point out that MAIN is not a test of how well children can recognize visual expressions. Rather, narratives elicited with MAIN can show how children infer and understand universal internal states of animate beings in carefully constructed, causally linked chains of events that are visually presented.

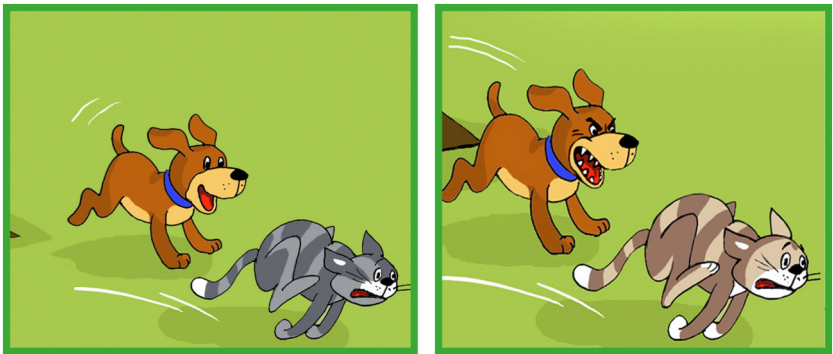
This is just one example of how the picture series were carefully revised numerous times before the best version to support the elicitation of appropriate ISTs was arrived at. All in all, over 200 variants were created by Lithuanian professional artist Loreta Valantiejienė through continuous dialogue with the team of MAIN developers before MAIN was launched as an open-access tool. Through its careful creation based on structured design, cross-cultural adaptability, and emphasis on clear depiction of protagonists’ internal states, MAIN provides a robust

framework for investigating ISTs as a core aspect of narrative competence across diverse linguistic backgrounds.

Previous Studies on ISTs in Narratives Elicited via MAIN

Since the launch of MAIN, numerous studies have used the tool to investigate the development of narrative skills in children aged between 5 and 9 years; however, only a few of these publications report on ISTs. Monolingual children in these studies spoke Dutch (Blom & Boerma, 2016; Boerma et al., 2016), Greek (Tsimpli et al., 2016), Finnish (Kunnari et al., 2016), Italian (Roch et al., 2016), and Luganda (Uziel et al., 2025). Bilingual children spoke Catalan–Spanish (Camus Torres & Aparici, 2025), English–Hebrew (Altman et al., 2016), Italian–English (Roch et al., 2016), Swedish–Finnish (Kunnari et al., 2016), Russian–Hebrew (Fichman et al., 2021, 2022), English–Swedish (Bohnacker, 2016), L2 Dutch with various L1s (Boerma et al., 2016), or L2 Greek with various L1s (Tsimpli et al., 2016).

Figure 3. Older version (left) versus final version (right) of the last picture of the Multilingual Assessment Instrument for Narratives Baby Birds story (Gagarina et al., 2012).



It is somewhat difficult to discern a common denominator in the research findings of these studies with respect to ISTs because not all studies performed their analyses according to the MAIN scoring protocols but instead developed systems of their own. Additionally, the studies investigate relatively small groups of children coming from diverse populations and do not use the same mode of story elicitation, which may have an effect on the results. For instance, two studies compare telling and retelling modes (Kunnari et al., 2016; Roch et al., 2016), three studies compare monolingual and bilingual children (Boerma et al., 2016; Kunnari et al., 2016; Tsimpli et al., 2016), six studies compare L1 and L2 in bilingual children (Altman et al., 2016; Bohnacker, 2016; Camus Torres & Aparici, 2025; Fichman et al., 2021, 2022; Kunnari et al., 2016; Roch et al., 2016), and five studies compare TD children and children with DLD. While some studies (Bohnacker, 2016; Roch et al., 2016) concentrate on *ISTs relevant for macrostructure*, that is, ISTs as initiating events and ISTs as reactions as described earlier (max six ISTs per story), other studies instead report on the frequency of *IST tokens*. Yet others focus on ToM-related versus non-ToM-related ISTs (Tsimpli et al., 2016) or on ISTs as enabling events (Fichman et al., 2021). Furthermore, ISTs were not an exclusive main topic for any of these studies but a secondary research interest in combination with story structure and story complexity.

Bearing in mind these limitations, there do appear to be some common trends. When different age groups such as preschoolers and primary school children were compared (Bohnacker, 2016; Camus Torres & Aparici, 2025; Roch et al., 2016), TD children generally showed a developmental increase with age in the diversity and frequency of ISTs—particularly perceptual terms such as *see* and emotional terms such as *happy* and *scared*. None of these studies investigated ISTs for age as a continuous variable. When IST production in monolinguals and bilinguals was compared, findings were inconclusive, as were the findings for ISTs in the bilinguals' L1 and L2. A clearer picture emerges when different elicitation modes were compared: IST production was higher in retelling as compared to telling (Kunnari et al., 2016; Roch et al., 2016). However, it remains to be shown whether this is truly an effect of elicitation mode effect (tell or retell) or type of story (Cat/Dog, Baby Birds/Baby Goats). When TD and DLD children were compared, the following trend was found: Monolingual and bilingual children with DLD produced fewer ISTs than TD children (Altman et al., 2016; Boerma et al., 2016; Fichman et al., 2021; Tsimpli et al., 2016).

Taken together, the findings on the use of ISTs in fictional narratives with MAIN are so far based on studies with relatively low numbers of participants aged 5–9 years. The results from these studies reveal a heterogeneous and

even contradictory picture: For instance, while some studies report more ISTs in L2 (e.g., Altman et al., 2016; Fichman et al., 2022), others observe more ISTs in L1 (e.g., Roch et al., 2016). Some studies show more use of IST (specifically, emotion terms) in bilinguals than in monolinguals (Boerma et al., 2016; Tsimpli et al., 2016). These inconsistencies underscore the need for a large-scale, cross-linguistic study for a wider age range to capture the full developmental trajectory of IST use in narratives in diverse linguistic and sociocultural contexts. This was the motivation for the present study.

The Present Study

Aims and Research Questions

The overarching aim of the study is to advance our understanding of children's oral narrative skills through the lens of IST development in fictional stories elicited with MAIN, using a large cross-linguistic data set. We begin by examining the overall developmental trajectory with age. Based on the existing literature (Berman & Slobin, 1994; Nicolopoulou & Richner, 2007; Pearson, 2002; Roch et al., 2016; Trabasso et al., 1992), we anticipate that the likelihood of producing ISTs increases as children grow older. However, this general trend is expected to be accompanied by variation. We then investigate which factors contribute to this variation within the broader pattern of age-related development in our sample. Specifically, we ask whether the developmental pattern of IST use is influenced by lingualism, defined here as the child's language status—monolingual or bilingual—and, in the case of bilinguals, their L1 and/or L2. Thus, we compare the developmental pattern of IST in monolingual and bilingual children and, for bilingual children, in their L1 versus L2. We do this for the entire data set and, additionally, to control for language, also for a subset of our data for which both monolingual and bilingual data were available for the *same* language. Effects of mono/bilingualism and L1/L2 are difficult to predict, as the results of earlier studies are inconclusive. We then investigate the effect of elicitation mode and ask whether and how IST production differs in telling and retelling. We anticipate that retelling will trigger more ISTs than telling, in line with the few existing studies (Kunnari et al., 2016; Roch et al., 2016). Moreover, we ask whether the distribution and production of ISTs is similar across the four MAIN stories and across IST types (initiating events vs. reactions). Finally, we ask how IST production interacts with story length measured by the total number of words (TNW). It is difficult to make any predictions here due to a lack of studies (but see Uziel et al., 2025). Our study is explorative in this regard; it is plausible though to expect measurable effects in a large sample such as ours. In summary, our research questions (RQs) are the following:

Table 3. The number, mean age (*SD*), and age range of participants by group.

Group	Number of children	Mean age in months (<i>SD</i>)	Age range in months
Monolingual	1,498	85.73 (27.40)	37–167
Bilingual	1,265	89.98 (28.67)	36–168
Total	2,763	88.27 (22.93)	36–168

RQ1: How does the production of ISTs in fictional stories elicited with MAIN develop in children ages 3–13 years?

RQ2: How is the production of ISTs influenced by age, lingualism (i.e., the child’s language status—monolingual or bilingual—and, in the case of bilinguals, their L1 and/or L2), and elicitation mode (telling, retelling)?

RQ3: Is there a difference in the production of ISTs as initiating events and reactions in relation to age, lingualism, and elicitation mode?

RQ4: Is there a difference in the production of ISTs between the four stories: *Cat*, *Dog*, *Baby Birds*, and *Baby Goats*?

RQ5: How does story length interact with the production of ISTs?

Method

Participants

A total of 2,763 monolingual and bilingual children ages 3;0–13;11 (years;months; $M_{\text{age}} = 88.27$ months, $SD = 22.93$) from 32 countries around the world, living in urban and rural areas, participated in the study.¹ They told or retold 5,507 narratives in 44 languages belonging to different language families (Afrikaans, Akan, Cantonese, Catalan, Coriglianese [Calabrian], Croatian, Danish, Dutch, English, Estonian, Farsi, Finnish, Georgian, German, Gondi, Greek, Halbi, Hebrew, Hindi, Irish Gaelic, Italian, Kannada, Kazakh, Korean, Lithuanian, Luganda, Malay, Mandarin, Polish, Russian, Scots Gaelic, Shugni, Slovak, Spanish, Swedish, Tagalog, Tajik, Turkish, Ukrainian, Urdu, Uyghur, Uzbek, Xhosa, Yakut; see Supplemental Material S1). Table 3 below presents more information about the distribution of participants across the monolingual and bilingual groups.

Children attended preschools or primary schools where the societal language is the language of schooling. By “monolinguals,” we mean children whose home language

matched the language of schooling and of society at large. “Bilinguals” were children who either grew up with two home languages (simultaneous bilinguals) or who were (pre)schooled in a language other than the home language (sequential bilinguals). L1 was operationalized as the language of the home, and L2 as the language of society if different from the home language.

Parents of all children provided written consent for participation in the study. In addition, parents, guardians, and teachers provided background information about the child, developmental language milestones, social status, and medical history. Based on this information, only TD children were included in the study (see Supplemental Material S1 for inclusion and exclusion information). In regions where it was safe to travel, fieldwork researchers systematically contacted nursery schools, kindergartens, and/or primary schools where the elicited languages are spoken, as well as community centers, heritage language teachers, and Saturday schools and parents, to recruit children. Letters, e-mails, and flyers were sent to principals and teachers, and/or telephone contacts were established. In oral and written communication with parents and institutions, researchers strove to use indigenous and heritage languages where relevant. The comprehensive recruitment effort led to a high rate of participant take-up (the respective response rates are given in Supplemental Material S1).

The data of our participants come from labs and individual (fieldwork) researchers around the world, who, at the initiative of the first author, joined forces to substantially advance our understanding of children’s acquisition of narratives. For the present article, MAIN data collected with the same materials and elicitation methods were combined into a large cross-linguistic data set and analyzed in the same way.

Materials and Elicitation

The MAIN elicitation tool (Gagarina et al., 2019) contains four stimuli sets of structurally parallel fictional picture sequences of six pictures each to elicit four stories with identical overall story structure and episode structure: *Cat*, *Dog*, *Baby Birds*, and *Baby Goats*. These are accompanied by a standardized protocol to assess both narrative production and comprehension. Since there are four

¹The large majority of children (89%) lived in urban regions.

picture sequences, bilingual children can be tested in both their languages with comparable stimuli without reusing the same story. This reduces training effects and boredom, which otherwise easily occur when the same material is used with the same child several times.

MAIN is administered to the child in person by a trained experimenter in a secluded space, and the entire procedure is audio-recorded (and/or video-recorded). The child is shown three envelopes containing picture sequences and asked to choose and open one envelope (what the child does not know is that they all contain the same story). The picture sequence is taken out and shown to the child but held away from the experimenter's view for nonshared visual attention. The child is reminded that the experimenter cannot see the pictures; this is done to encourage the child to tell the story as explicitly as possible and to discourage the use of deictic gestures instead of words.

In this study, MAIN is administered in two different modes: telling and retelling. In the telling mode, the child generates the story (typically *Baby Birds* or *Baby Goats*) without any prior model provided by the adult. For familiarization, the entire picture sequence is first shown to the child and then folded up again. Then, the experimenter opens the pictures two by two and lets the child tell the story until the entire picture sequence is unfolded. In the retelling mode, the child hears the story (typically *Cat* or *Dog*) from an adult (or prerecorded via headphones) while looking at the pictures and then tells the same story to another experimenter. The pictures are visible to the child throughout the test to minimize memory demands.

Authentic examples of a child telling the *Baby Birds* story and retelling the *Cat* story are given in narratives (1) and (2), with English translations underneath. ISTs produced by the child are italicized, and ISTs relevant for macrostructure targeted in MAIN are shown in bold. It is the ISTs relevant for macrostructure (maximally six per story) that are investigated in the present study. Please note that the total number of IST tokens produced by the child in a story can be higher than the number of ISTs relevant for macrostructure targeted in MAIN.

(1) Swedish-speaking 7-year-old telling the *Baby Birds* story:

det va(r) två små ungar som hade kläcks från sina nya ägg
och då ska mamman flyga iväg o(ch) hämta mat
o(ch) då kom de(t) en *hungrig* katt
o(ch) då så klättra(r) han upp i &kl [/] trädet
o(ch) då när mamman kommer tillbaka o(ch) ger mask till sin första unge
o(ch) då < tar katten den and(ra) > [/] den andra ungen

dock *ser* en vo(vve) [/] hund de(t) o(ch) biter katten i svansen o(ch) drar ner den.

sen jagar han efter den där katten

och < fågeln blir> [/] mamman blir *jätterädd* för katten

o(ch) sen blir hon *så glad* att hunden jagar bort den där dumma lilla katten.

“There were two small babies who had hatched from their new eggs. And then the mother is going to fly off to get some food. And then came a *hungry* cat. And then he climbs up in xx the tree. And then when the mother comes back and gives a worm to her first baby. And then the cat takes the other, the other baby. But a woof-woof, a dog *sees* this and bites the cat's tail and pulls it down. Then he chases after that cat. And the bird, the mum gets *super scared* of the cat. And then she gets *so happy* that the dog chases away that stupid cat.”

Story length (TNW): 103; IST tokens: 4, with two ISTs as initiating events (*hungrig* “hungry,” *ser* “see”) and one IST as a reaction (*så glad* “so happy”).

(2) Swedish-speaking 7-year-old retelling the *Cat* story:

de(t) var en gång en katt som *såg* en fjäril i en buske

<och sen > [/] och sen hoppa(de) katten upp för att fånga fjärilen

o(ch) så kom en pojke me(d) en hink full med fiskar från en fisketur

han &ehm [/] alltså han *såg* hur katten sprang efter den där fjärilen

fast hon var inte snabb nog o(ch) fjärilen kom undan o(ch) hon ramlade in i busken

pojken *hajade till* o(ch) tappa(de) sin boll o(ch) den åkte ner i sjön

han *tänkte*: “*åh nej*, min boll!”

men sen så var han &ehm alltså han försökte fiska upp den med fiskespöt

o(ch) så fick han upp den o(ch) blev *jätteglad*, fast han *såg* inte va(d) katten höll på me(d)

hon *såg* fiskarna i hinken o(ch) *tänkte mums*

så hon mumsa(de) i sig alla fiskarna och pojken *märkte* inte de(t).

“Once there was a cat that *saw* a butterfly in a bush. And then, and then the cat jump(ed) to catch the butterfly. And then a boy came with a bucket full of fish from a fishing-trip. He uhm well he *saw* how the

cat ran after that butterfly. But she wasn't fast enough and the butterfly got away and she fell into the bush. The boy got *startled* and lost his ball and it went down into the lake. He *thought oh no*, my ball! But then he was uhm well he tried to fish it up with his fishing rod. And then he got it out and was *super happy*, but he didn't *see* what the cat was doing. She *saw* the fish in the bucket and *thought yummy!* And she munched all the fish and the boy didn't *notice*."

Story length (TNW): 136; IST tokens: 8, with three ISTs as initiating events (*såg* "saw," *tänkte åh nej* "thought oh no!", *såg* "saw"/*tänkte mums* "thought yummy!") and one IST as a reaction (*jätteglad* "super happy").

Data

Each lab (research team) was responsible for data collection, transcription, and IST coding according to the MAIN scoring sheets (Gagarina et al., 2012, 2019) and for the calculation of story length measured in number of words for each narrative. An overview of the numbers of narratives broken down by group (monolinguals, bilinguals, and, in the latter, L1 and L2), mode (telling, retelling), and story is given in Tables 4 and 5.

Children always told and retold different stories. The number of narratives told and retold and the number of the different MAIN stories is not fully balanced for the sample as a whole ($N = 5,507$) across the 45 labs. However, in each individual lab, administration of the stories was counterbalanced according to the MAIN manual (Gagarina et al., 2019: 5). The labs also supplied background information about their participants and recruitment strategy, which was then collated into one file by the first two authors (see Supplemental Material S1).

Data Treatment and Statistical Analyses

Audio (and in some cases video) recordings were transcribed manually by trained native speakers within the respective research teams. For all narratives, length was calculated as TNW, excluding mazes (repetitions, reformulations). The transcripts were searched manually for ISTs and coded following the MAIN scoring sheets. These have been adapted to the respective language following the adaptation guidelines (Bohnacker & Gagarina, 2019, pp. iv–xii). Many of the experts who made these adaptations are also co-authors of the present article. IST coding was performed by trained native speakers of the respective language, coached by the first two authors and other experienced members of their teams. Each lab ensured and confirmed their own intercoder reliability. Based on a template provided by the first author, the different labs entered their coded data into one master table in Excel format, which

was then carefully checked and cleaned in cooperation between the first author and her team and the respective labs. The resulting data set of 5,507 narratives forms the basis for the statistical analyses of the 2,427 ISTs as initiating events or reactions. As outlined above, a narrative may contain up to six such ISTs (three ISTs as initiating events and three ISTs as reactions).

In the following, we do not analyze the actual occurrence of ISTs but rather use more powerful statistics to investigate the *likelihood* (probability) of a child producing ISTs (as initiating events or as reactions). For these analyses, we fitted a series of generalized linear mixed models, implemented in a Bayesian framework in R using the *brms* package (Bürkner, 2017). The dependent variable was modeled as a binomial distribution (unless otherwise stated) so that we estimated the probability that a child would produce ISTs. We used default priors implemented in the package for all parameters. To compare models, we used Bayesian approximate leave-one-out cross-validation (Vehtari et al., 2017) based on differences in expected log posterior density (ELPD) estimates and the associated standard error (*SE*). Based on suggestions in the literature (Sivula et al., 2020), we consider a model to be substantially better when the ELPD in favor of that model exceeds 2 times the *SE* of the difference. To evaluate individual predictors in a model, we investigated its posterior mean and the associated 95% credible interval (CrI). We classified a predictor to be substantial when its 95% CrI did not overlap with 0. Numerical predictors were scaled to have a mean of 0 and an *SD* of 1.

Results

ISTs and Age

First, we asked how the probability of producing ISTs changes with age. We included all available data (5,507 narratives, 2,763 children, 44 languages). The only fixed effect in the model was age; random effects comprised an intercept for child and test language and a slope for age within test language. As shown in Figure 4, the probability of producing ISTs increased substantially with age (fixed effect in Bayesian regression model: $\beta = .71$, 95% CrI [0.52, 0.91]). Given that age has such a strong impact, we included it as a predictor in all further analyses.

IST Influenced by Age, Monolingualism/Bilingualism, L1 and L2, and Elicitation Mode

Second, we asked whether—in addition to age—the probability of producing ISTs is influenced by elicitation mode (telling vs. retelling), language group (mono- vs.

Table 4. Number of narratives by group and elicitation mode.

Group	Telling	Retelling	Total
Monolingual	1,387	744	2,131
Bilingual L1	967	664	1,631
Bilingual L2	952	793	1,745
Total	3,306	2,201	5,507

Note. L1 = first language; L2 = second language.

bilingual), and, in bilinguals, their L1 versus L2. We used all the available data (5,507 narratives from 2,763 children in 44 languages). The best fitting model included as fixed effects main effects of age, language group, elicitation mode, and L1 versus L2 in bilinguals. As random effects, the model included an intercept for child and language and a slope for age within language. Across languages, children improved with age (fixed effect in Bayesian regression model: $\beta = .68$, 95% CrI [0.47, 0.90]). Numerically, bilinguals were more likely to produce ISTs; however, this predictor was not reliably different from zero ($\beta = -.09$, 95% CrI [0.07, -0.23]). Children were more likely to produce ISTs when retelling the story compared to when telling it ($\beta = -1.11$, 95% CrI [-1.17 , -1.05]). Being tested in L1 versus L2 had no substantial effect ($\beta = -.01$, 95% CrI [-0.16 , 0.14]; see Figure 5A).

Note, however, that the model described above fitted the data only marginally better compared to two other models. The second-best model (ELPD difference = -1.10 , *SE* of difference = 3.01) included an additional interaction between age and language group, which was reliably different from zero ($\beta = -.27$, 95% CrI [-0.45 , -0.08]). This points to different developmental trajectories for monolinguals and bilinguals in that bilinguals showed a steeper increase in performance with age. A third model, which did not include language group as a predictor, fit the data slightly worse (ELPD difference = -2.55 , *SE* of difference = 1.91). Taken together, this pattern of results suggests that the parameter of language group does not improve predictions substantially. Furthermore, if anything, bilinguals have an advantage over monolinguals.

For the remaining analysis, we will include language group as a main effect.

To compare monolingual and bilingual performance more directly, we refit the same models on a subset of the data for those languages for which we had data from monolinguals as well as bilinguals (1,403 narratives from 945 children in 13 languages). The best fitting model for this subset included an interaction between age and language group, while the second best model—which was only marginally worse—did not include language group at all (ELPD difference = -6.88 , *SE* of difference = 4.6). In the interaction model, the predictor for the interaction term did not include zero ($\beta = -.39$, 95% CrI [-0.55 , -0.23]; see Figure 5B). Also in this model, that is, when the two groups are matched for languages, the probability of the bilinguals' trajectory showed a steeper increase than that of the monolinguals in performance with age (see Figure 5B).

ISTs as Initiating Events or Reactions

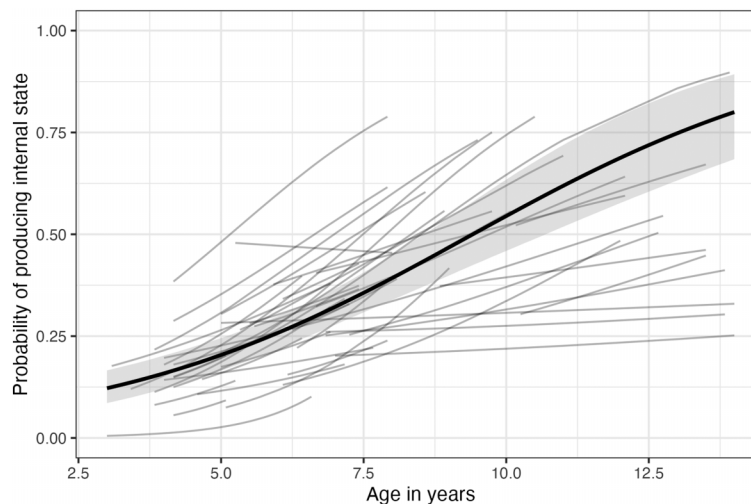
The third RQ asked whether the production of ISTs as initiating events was more likely than that of ISTs as reactions, in relation to age, lingualism (mono- vs. bilingual and, in bilinguals, their L1 vs. L2), and elicitation mode. We used all the available data. The model had the same general structure as the RQ2 model with the added predictor IST type (initiating event vs. reaction), which was included as a main effect, an interaction with age and a random slope within language. Children were more likely to produce ISTs as initiating events than as reactions ($\beta = -1.22$, 95% CrI [-1.54 , -0.91]). This pattern became even more pronounced with age (interaction between age and IST type: $\beta = -.34$, 95% CrI [-0.42 , -0.25]; see Figure 6). As in the previous models, children were more likely to produce ISTs when retelling the story compared to when telling it ($\beta = -1.22$, 95% CrI [-1.28 , -1.15]). Being tested in L2 versus L1 had no substantial effect ($\beta = .00$, 95% CrI [-0.16 , 0.16]). Monolinguals and bilinguals did not differ ($\beta = -.10$, 95% CrI [-0.24 , 0.04]; see Figure 6).

Table 5. Number of narratives by story, group, and elicitation mode.

Group	Telling				Retelling				Total
	Cat	Dog	Baby Birds	Baby Goats	Cat	Dog	Baby Birds	Baby Goats	
Monolingual	636	133	312	306	663	76	1	4	2,131
Bilingual L1	93	81	369	424	303	300	29	32	1,631
Bilingual L2	66	69	426	391	316	268	98	111	1,745
Total	795	283	1,107	1,121	1,282	644	128	147	5,507

Note. L1 = first language; L2 = second language.

Figure 4. Model predictions for average developmental trajectory. The saturated line shows the trajectory across languages with 95% credible interval. The transparent lines show model predictions for the different test languages. The length of each line shows the range for which there are data in any of the languages to inform the model estimates. L1 = first language; L2 = second language.



ISTs and Story Type

The fourth RQ focused on the different stories that are shown in the pictures (*Cat, Dog, Baby Birds, Baby Goats*) and asked whether the production of ISTs was more likely for some stories compared to others. We included all the available data. Once again, the model had the same general structure as the RQ2 model with the added predictor story (*Cat, Dog, Baby Birds, Baby Goats*), which was included as a main effect and a random slope within language. A model including an interaction between age and story did not fit the data better (ELPD difference = -3.01 , SE of difference = 1.92). The *Cat* story served as the reference story. In comparison to the *Cat* story, children were more likely to produce ISTs for the *Baby Goats* ($\beta = .57$, 95% CrI [$0.32, 0.81$]) and *Baby Birds* ($\beta = .39$, 95% CrI [$0.15, 0.63$]) stories. There was no difference between the *Cat* and the *Dog* story ($\beta = -.39$, 95% CrI [$-0.64, -0.13$]; see Figure 7).² Once again, children were more likely to produce ISTs when retelling the story compared to when telling it ($\beta = -1.44$, 95% CrI [$-1.52, -1.35$]). Being tested in L1 versus L2 had no substantial effect ($\beta = .03$, 95% CrI [$-0.13, 0.18$]). Monolinguals and bilinguals did not differ ($\beta = -.07$, 95% CrI [$-0.13, 0.18$]; see Figure 7).

ISTs and Story Length

Finally, we asked whether the length of the story influenced the probability of producing ISTs. For this, we

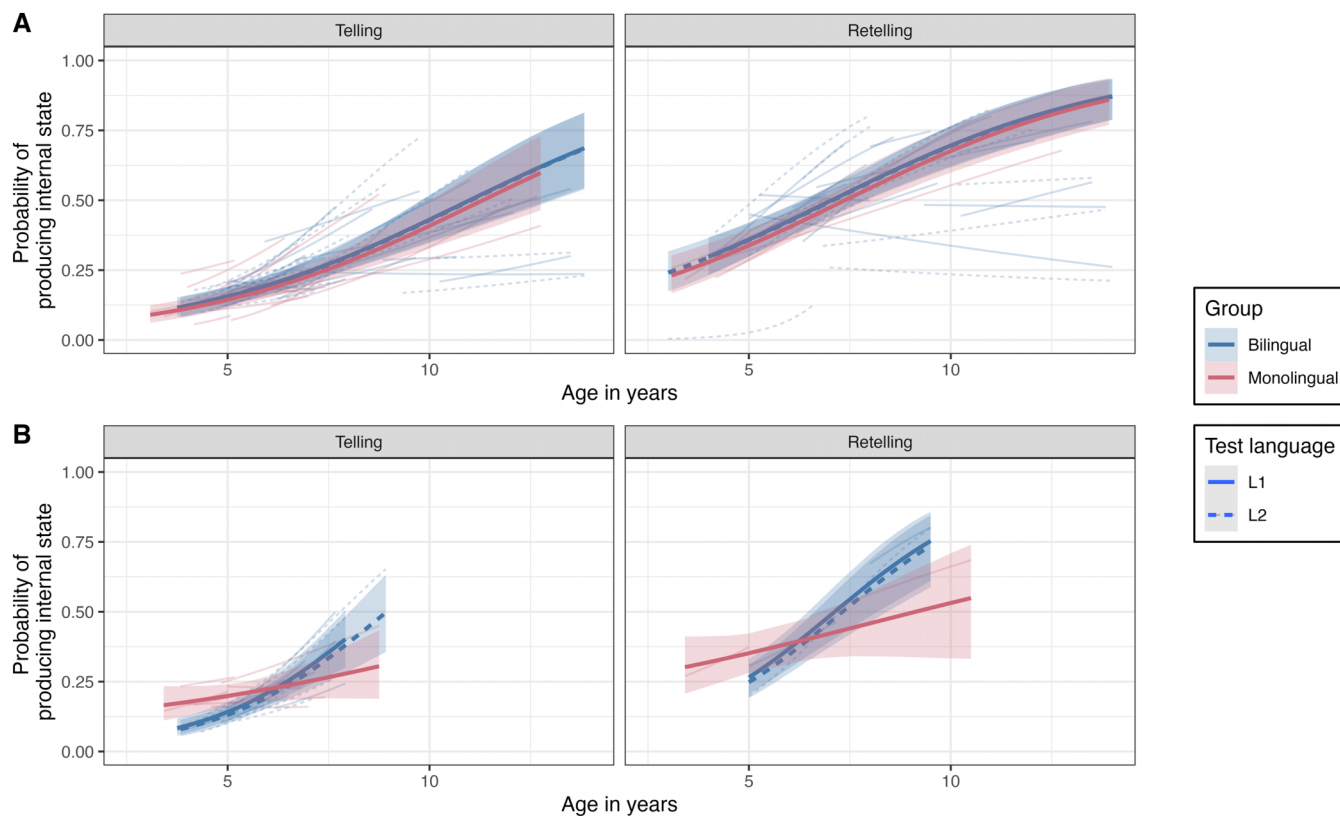
²The Appendix includes a model where the effect of story type is analyzed for each language separately.

added story length, operationalized as the TNW, as an additional predictor to the model described in RQ2. Information about story length was only available for a subset of children, although this subset was still very large (4,183 narratives from 2,189 children in 35 languages).

We fitted two types of models that treated story length differently. In the first model, story length was scaled across languages so that we directly compared the number of tokens produced *irrespective* of the language. For the second model, story length was scaled *within* language and thereby accounted for the effect that languages might be more or less “wordy.” By this, we mean that, depending on the linguistic structure of a language, the number of words in an average narrative may differ. For instance, agglutinative languages such as Turkish, Korean, and Finnish may use fewer (but longer) words (with much morphological inflection) to express the same story content when compared to analytic languages such as English and Chinese (that will use more words but less inflected ones). For our statistical model, this means that the predictors show how the probability of producing an IST changes when producing a story longer (or shorter) relative to all stories produced in that language. In both models, we included a main effect of story length, an interaction with age, and a random slope within test language.

In the first model, story length had a substantial effect on the probability of producing ISTs in that longer stories led to more ISTs ($\beta = .49$, 95% CrI [$0.39, 0.59$]). However, this effect decreased with age (interaction between age and story length: $\beta = -.16$, 95% CrI [$-0.22, -0.11$]), suggesting that producing a longer story does not increase the probability of producing an IST for older

Figure 5. (A) Model predictions for developmental trajectories based on the fixed-effects model for all data. The two facets show trajectories for the two elicitation modes (telling and retelling). The saturated lines show trajectories across languages with 95% credible interval. The transparent lines show model predictions for the different languages. The length of each line shows the range for which there are data in any of the languages to inform the model estimates. Color denotes the language group (mono- vs. bilinguals) and the line type (solid vs. dashed) whether bilinguals were tested in their first language (L1) or second language (L2). (B) The same visualization but for the interaction model based on data for languages for which there were data for the same language in mono- and bilinguals.



children (see Figure 8). In the second model, these effects were similar: main effect of story length ($\beta = .36$, 95% CrI [0.28, 0.43]) and negative interaction ($\beta = -.14$, 95% CrI [-0.19, -0.09]; see Figure 8). All other effects were largely similar to the model for RQ2 (here only reported for the first model): Children were more likely to produce ISTs when retelling the story compared to when telling it ($\beta = -.97$, 95% CrI [-1.04, -0.89]). Being tested in L2 versus L1 had no substantial effect ($\beta = -.14$, 95% CrI [-0.32, 0.04]). Monolinguals and bilinguals did not differ ($\beta = -.03$, 95% CrI [-0.17, 0.11]; see Figure 8).

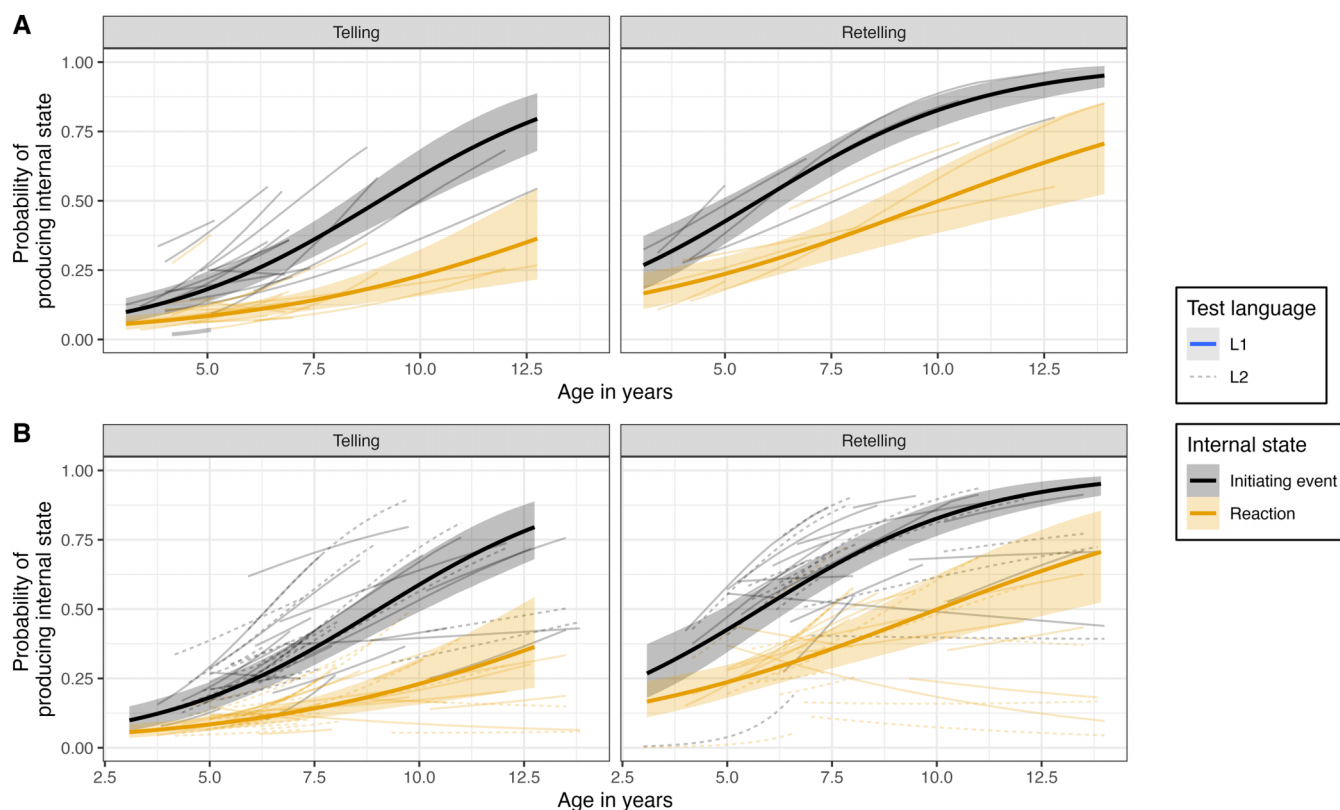
Discussion

This article investigated the development of narrative skills in children ages 3–13 years through the lens of ISTs in fictional narratives elicited with MAIN. Overall, we found a clear effect of age over lingualism, elicitation

mode, and story type. This is, of course, as expected but has never been shown for such a large and linguistically varied data set and within such a big age range. The finding that the probability of producing ISTs increases with age is in line not only with the results of earlier studies that compared two age groups for much smaller samples elicited via MAIN (Bohnacker, 2016; Roch et al., 2016; Uziel et al., 2025) but also with studies using different elicitation methods and materials (e.g., Berman & Slobin, 1994; Pearson, 2002; Stein, 1988; Stein & Glenn, 1979). The finding concerning age can be considered overarching: It goes beyond existing studies and covers 10 years of children's development, using data elicited with the same instrument and analyzed according to a uniform protocol.

We divided all participants into groups based on lingualism (monolinguals vs. bilinguals and, in bilinguals, their L1 vs. L2) and compared the production of their ISTs in the two elicitation modes—telling and retelling for

Figure 6. Model predictions for developmental trajectories. The facets show trajectories for the two elicitation modes (telling and retelling) and for language groups (mono- vs. bilinguals). The saturated lines show trajectories across languages with 95% credible interval. The transparent lines show model predictions for the different languages. The length of each line shows the range for which there are data in any of the languages to inform the model estimates. Color denotes the internal state term type (initiating event vs. reaction); line type (solid vs. dashed) shows whether bilinguals were tested in their first language (L1) or second language (L2).

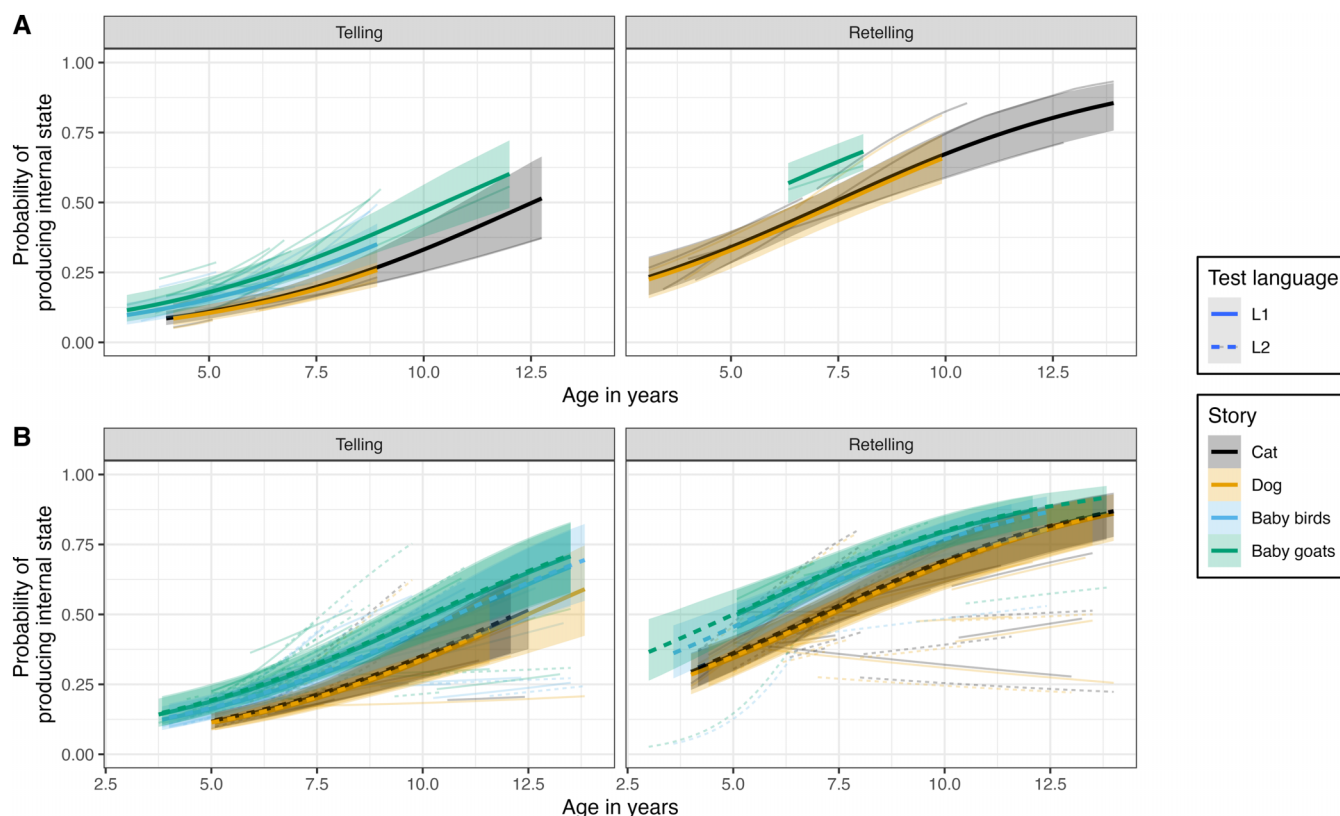


all stories. Our results show that the probability of producing ISTs is consistently higher for retelling than telling, echoing results of previous smaller scale studies (Fiestas & Peña, 2004; Kunnari et al., 2016; Roch et al., 2016). This is not particularly surprising, as in the retelling mode the children will have heard the internal states of the protagonists mentioned in the story they listened to and are asked to retell. They may thus profit from the modeled story and the lexical “priming” of ISTs. Interestingly, however, the probability of producing ISTs relevant for macrostructure was found to be similar for monolinguals and bilinguals and also similar for the bilingual children’s L1 and L2. If the development of ISTs in bilinguals is more strongly influenced by age (aka cognitive maturity) than by lingualism, as the present findings suggest, this would be in line with studies of other aspects of narrative macrostructure (story components, episodic structure) where bilingual children (simultaneous and sequential) have been found to exhibit similar complexity levels in their two languages, L1 and L2 (Bohnacker et al., 2022; Fiestas & Peña, 2004; Gagarina et al., 2015; Lindgren & Bohnacker,

2020; Pearson, 2002). A common way to explain such findings has been to assume that bilingual children at a given age have a similar understanding of story events in their two languages and are able to transfer conceptual narrative abilities from one language to the other (Berman, 2001; Paradis et al., 2021, pp. 170–171; Pearson, 2002). This could likewise apply to ISTs as initiating events and reactions, that is, ISTs relevant for macrostructure and to ISTs as initiating events building the full episode together with attempts and outcomes (Gagarina et al., 2021). However, since we did not have objective measures of language proficiency for all of our participants (such as tests of vocabulary, fluency, comprehension), we cannot exclude the possibility that the absence of differences between the L1 and L2 in bilinguals in our results could reflect heterogeneity in proficiency rather than a true symmetry in the use of ISTs across languages.

Then, we examined the production of ISTs as initiating events versus reactions, in relation to three key factors: age, lingualism (monolinguals vs. bilinguals and, in

Figure 7. Model predictions for developmental trajectories. The facets show trajectories for the two elicitation modes (telling and retelling) and for language groups (mono- vs. bilinguals). The saturated lines show trajectories across languages with 95% credible interval. The transparent lines show model predictions for the different languages. The length of each line shows the range for which there are data in any of the languages to inform the model estimates. Color denotes the story (*Cat*, *Dog*, *Baby Birds*, *Baby Goats*); line type (solid vs. dashed) shows whether bilinguals were tested in their first language (L1) or second language (L2).

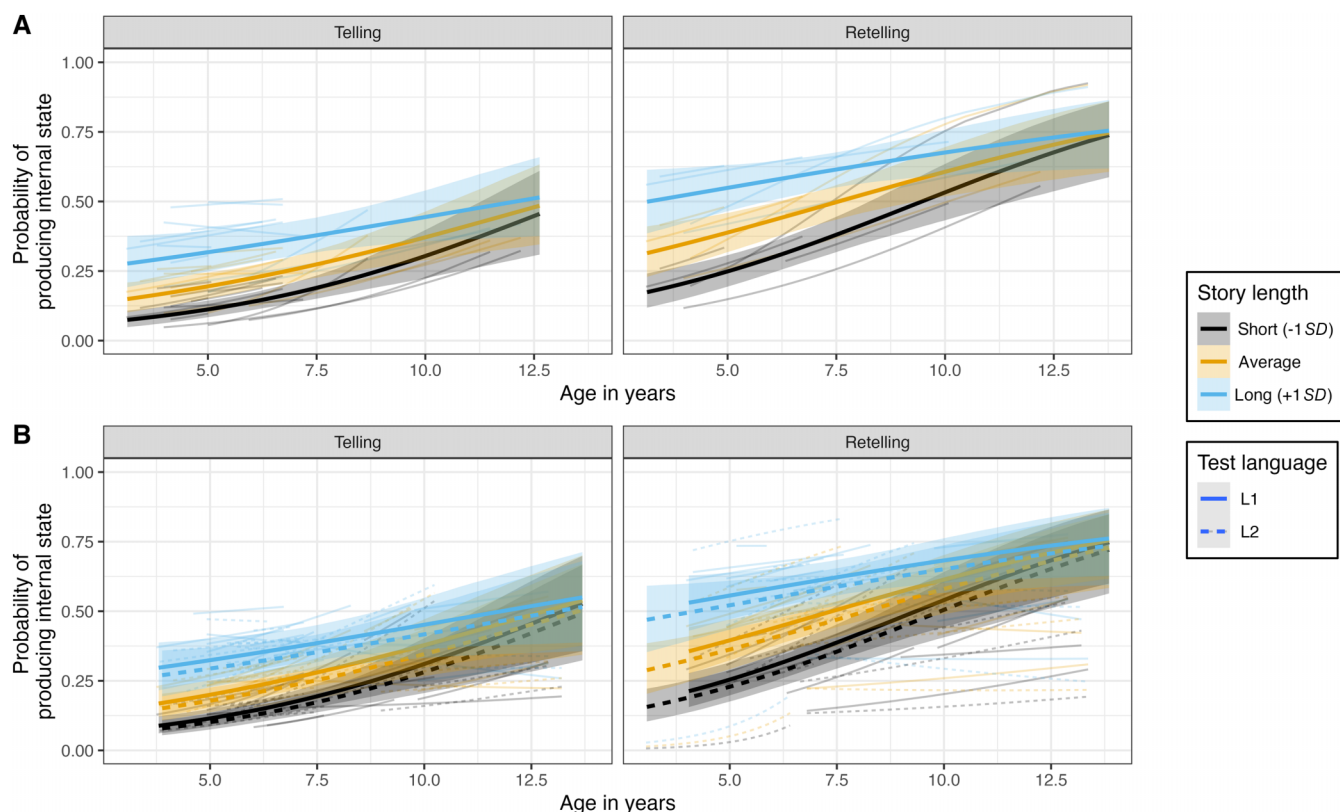


bilinguals, their L1 vs. L2), and elicitation mode (telling vs. retelling). Children at younger ages were generally unlikely to include ISTs in their narratives, though they were more likely to do so for ISTs as initiating events than as reactions, and this difference between the two types increased with age, especially for telling (see Figure 6). The likelihood of producing internal states as initiating events—as opposed to reactions—was higher across all ages and language groups (monolinguals and bilinguals in both L1 and L2), irrespective of the elicitation mode (telling or retelling). This robust finding highlights the prominence of ISTs as initiating events as triggers for narrative episodes and overall plot structure. The finding cannot be directly compared to other studies investigating the use of ISTs in elicited fictional narration. To our knowledge, there are no studies based on theoretically grounded and visually represented episodic structure in which ISTs function as initiating events and reactions within a narrative as in the MAIN scheme.

Differences in the production of ISTs between the four stories, *Cat*, *Dog*, *Baby Birds*, and *Baby Goats*, were

investigated. The results indicate that the stories *Cat* and *Dog* pair together, while *Baby Birds* and *Baby Goats* form another pair, revealing two distinct trends. In the *Baby Birds* and *Baby Goats* pair, the probability of IST production is consistently higher than in the *Cat* and *Dog* stories. We suggest that this difference can be attributed to three key factors: (a) the *Baby* stories feature a greater number of animate protagonists engaged in caretaker–child relationships; (b) their plots involve life-threatening situations (e.g., hunger, the threat of being eaten) and heightened emotional intensity; and (c) the episodes are more intricately interwoven, contributing to a more dramatic narrative structure. These results have implications for future work. If the goal of narrative elicitation is to obtain a higher number of ISTs, visual materials should be used that preferably depict characters involved in events connected to emotionally salient or potentially threatening situations (within ethical limits), rather than neutral activities. In addition, the materials should include episodes that feature animate protagonists not only as agents but also as affected participants or patients, as such scenarios are

Figure 8. Model predictions for developmental trajectories. The facets show trajectories for the two elicitation modes (telling and retelling) and for language groups (mono- vs. bilinguals). The saturated lines show trajectories across languages with 95% credible interval. The transparent lines show model predictions for the different languages. The length of each line shows the range for which there are data in any of the languages to inform the model estimates. Color denotes story length and the line type (solid vs. dashed) whether bilinguals were tested in their first language (L1) or second language (L2).



more likely to elicit ISTs due to reciprocity, for example, babies being hungry or in danger and parents being worried. Overall, vivid and engaging events appear to be particularly effective for IST elicitation.

Finally, we examined the relationship between the production of ISTs and story length (measured by the total number of word tokens). For younger children, there was a positive association with story length, but not for older children. As shown in Figure 8, this suggests that younger children exhibit much more variation in story length than older children; that is, length not only increases substantially with age but also becomes more uniform. Very short stories in younger children only include the bare bones of macrostructure (Berman & Slobin, 1994; Pearson, 2002), and ISTs may not be part of this. For instance, if some character and/or some events in the stories are omitted and not mentioned by the child, ISTs referring to the character's internal state as initiating events or reactions will not be verbalized either. With increasing length and story detail, there are more options for the child to produce

ISTs. However, beyond a certain story length, there is no further advantage in increasing length with regard to the well-formedness of the narrative and the production of ISTs relevant for macrostructure. Future work could pursue the production of ISTs in relation to other measures of language productivity than story length measured in TNW (e.g., number of different words, mean length of utterance or communication unit).

Limitations

This study has several limitations. It was not possible to perfectly match the data sets from all labs. The number of children is not identical across languages, elicitation modes, groups, and ages (e.g., we do not have *Cat* or *Dog* telling for children ages 4–5 years for all languages). As for data elicitation and analysis, not all labs/authors were able to participate in regular joint meetings to discuss language-specific or ambiguous scoring cases, even though we did our best to ensure that all adaptations followed the established guidelines and that analyses were conducted according to

the scoring protocols. Finally, language proficiency had not been measured in all labs, so we cannot exclude the possibility that the absence of differences in the use of ISTs between the L1 and the L2 in bilinguals could also reflect heterogeneity in language proficiency.

Conclusions

This study advances our understanding of narrative skills in several important ways. First, the production of fictional narratives appears to be strongly related to age. Using ISTs is a skill that develops over the course of many years in children: Descriptive observation of the probability slope indicates a pattern of gradual change from ages 3 to 5 years, with a somewhat steeper increase between ages 6 and 10 years and slower growth thereafter. Second, the finding that the probability of IST production does not differ between monolinguals and bilinguals—and, within bilinguals, between their L1 and L2—underscores the cognitively driven nature of IST use. This suggests that the ability to infer and express internal states is less dependent on language proficiency and more rooted in general cognitive development. Such a conclusion is further supported by the observation that, with increasing age, IST production becomes less dependent on narrative length, indicating a shift toward more efficient and conceptually grounded use of internal state language.

Third, the data show that ISTs are much more likely to be produced as initiating events rather than as reactions, highlighting their function as triggers of an episode and drivers of plot development. This finding reinforces the conceptual importance of ISTs as foundational elements in the development of children's narrative skills. Fourth, earlier research on fictional narratives (using other tools than MAIN) found that young children rarely mentioned the ISTs of story characters (Berman & Slobin, 1994; Nicolopoulou & Richner, 2007; Stein & Glenn, 1979), and this was also found in the present study. Nevertheless, MAIN is an effective tool for investigating ISTs as a key component of children's narrative skills, as there were the same age-related developmental trends irrespective of elicitation mode and story type. Regarding story type, *Baby Birds* and *Baby Goats* paired together and showed higher likelihoods for ISTs to be produced than the *Cat* and *Dog* stories. We attributed this to the *Baby Birds*' and *Baby Goats*' dramatic content (which includes more characters, an attacker and a rescuer, and matters of life and death). Researchers and clinicians interested in children's verbalization of ISTs may therefore want to preferably use the *Baby Birds* and *Baby Goats* stories for elicitation.

Finally, the consistent finding that retelling is more likely to elicit the production of ISTs—regardless of age,

story type, or language background—points to a robust facilitative effect of this elicitation mode. We suggest that retelling can be used with children whose language and cognitive skills are not yet fully developed or impaired, since it supports cognitive–linguistic integration in narrative production, promoting deeper engagement with characters' internal states.

Clinical Implications

MAIN is a freely available task that speech-language pathologists and therapists working with bilingual children can easily add to their toolkit. The present study has shown that MAIN is a suitable task to evaluate IST production in narratives (see also Winters et al., 2022). Although this study did not include a sample of children with DLD, MAIN has been used in previous research to differentiate children with typical development from those with DLD, and the use of ISTs in MAIN has been found to differ between these two groups of children (e.g., Boerma et al., 2016; Fichman & Altman, 2024; Kraljević et al., 2020; Mazlan et al., 2025; Tsimpli et al., 2016; see more on <https://main.leibniz-zas.de/en/research/publications/>). Against this background, the results of the current study offer several implications for the assessment and intervention of children's oral narrative skills through the lens of ISTs. Internal states, which are inferred rather than directly observable, and in particular their linguistic encoding, provide a window into both language and cognition, linking these domains in a meaningful way and underscoring their relevance for clinical practice. This interpretation is further supported by other contributions in this clinical forum. The suitability of MAIN for capturing such information is evident, and on this basis, we derive the following implications.

While children's use of ISTs continued to increase over the course of several years, ISTs as initiating events had much higher likelihoods than ISTs as reactions to be produced by TD children. Consequently, if it becomes clear during clinical assessment that a child has difficulties producing ISTs as initiating events, these may constitute an appropriate target for intervention, considering how important ISTs are for driving the narrative plot. Retelling was found to consistently boost IST use (across all language groups and stories), which suggests that this elicitation mode can serve as a valuable model or scaffold in clinical practice. The *Baby Birds* and *Baby Goats* stories may be particularly suitable in this regard, as they were found to have a higher probability of ISTs being produced by the children (irrespective of language group and elicitation mode). Finally, since there are four parallel versions in MAIN, they can be used for dynamic assessment (Gillam et al., 1999; Peña et al., 2006; Petersen et al., 2024) and

progress monitoring in a child, as well as for assessment or intervention in a bilingual child's two languages.

Ethics Statement

The research reported here complies with the ethical regulations of the respective countries and was carried out in accordance with the principles of the revised Helsinki Declaration. For details on how this was organized in each country/district, please see the Supplementary material. The parents of all the participating children provided informed consent and the children assented.

Data Availability Statement

All elicitation materials, including guidelines, scoring sheets, and analysis protocols for each of the 44 languages, are openly accessible after registration via the Leibniz-ZAS repository: <https://www.leibniz-zas.de/index.php?id=964>. Where parental consent permits, audio recordings and narrative transcriptions will be made available through CHILDES, a widely used open database for child language research: <https://talkbank.org/childes/>. Additionally, portions of the raw data—including the Leibniz ZAS corpus of MAIN (Version 0.1.0) by Topaj, Rizaeva, Sternharz, and Gagarina et al. (2021)—have been published on Zenodo: <https://doi.org/10.5281/zenodo.4724970>. The data and code to reproduce the analysis can be found in the following repository (https://github.com/manuelbohn/ist_narratives) and on the Zenodo version of the repository (<https://doi.org/10.5281/zenodo.18599609>).

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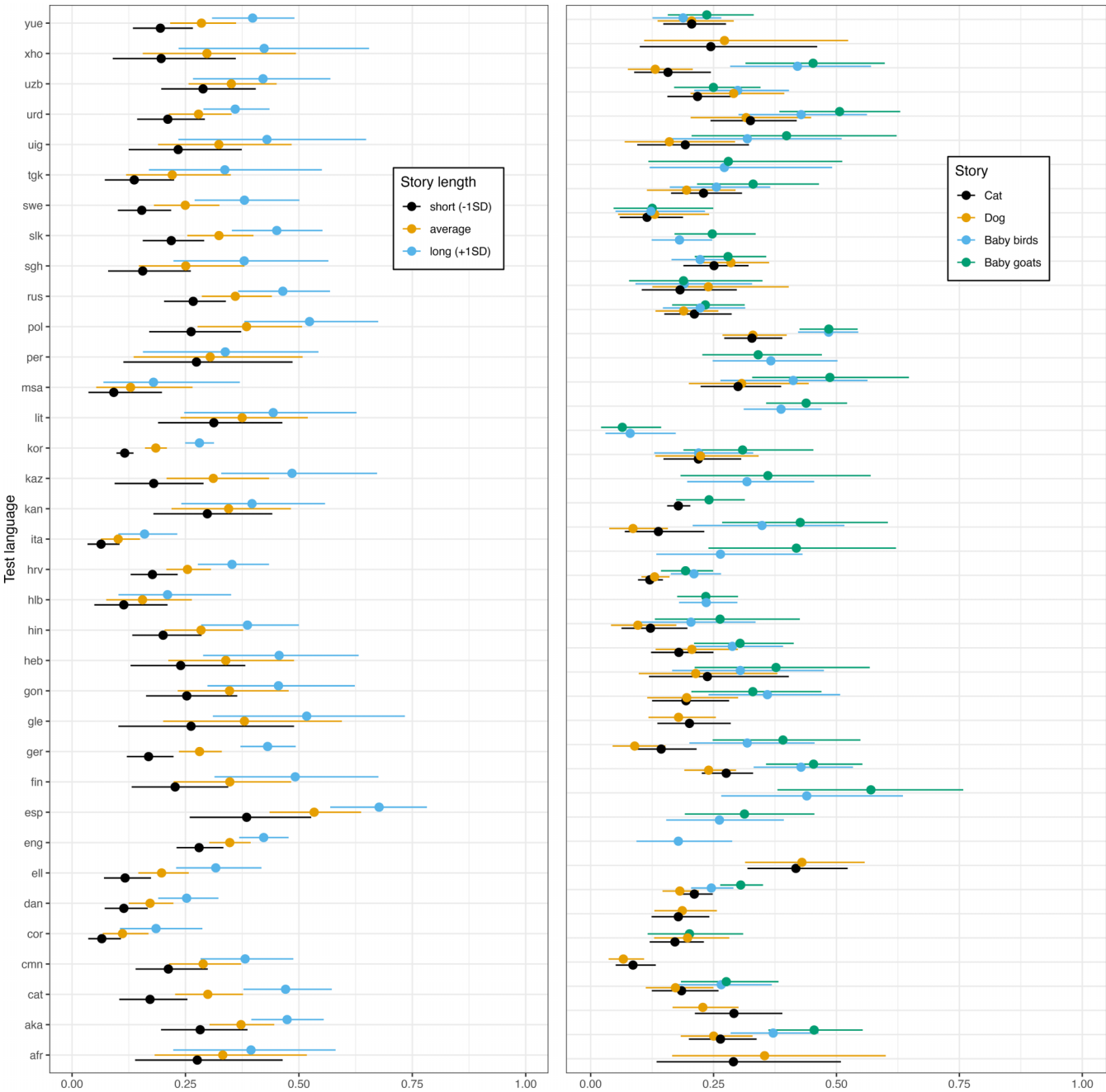
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Appendix

Probability of IST Production per Language



Appendix Figure: Model predictions for IST production by language. Facets show predictions from different models including varying focal predictors (left: story length; right: story). Points show posterior means with 95% CrI. Color denotes the levels of the focal predictors. Predictions are for bilingual children, tested in their L1, at the average age and in the telling condition. Predictions are plotted on the x-axis, and languages are plotted on the y-axis: Afrikaans [afr], Akan [aka], Catalan [cat], Mandarin [cmn], Coriglianese (Calabrian) [cor], Danish [dan], Greek [ell], English [eng], Spanish [esp], Finnish [fin], German [ger], Irish Gaelic [gle], Gondi [gon], Hebrew [heb], Hindi [hin], Halbi [hlb], Croatian [hrv], Italian [ita], Kannada [kan], Kazakh [kaz], Korean [kor], Lithuanian [lit], Malay [msa], Farsi [per], Polish [pol], Russian [rus], Shugni [sgh], Slovak [slk], Swedish [swe], Tajik [tgk], Uyghur [uig], Urdu [urd], Uzbek [uzb], Xhosa [xho], Cantonese [yue]. No story length measures were available for the following languages: Dutch, Estonian, Georgian, Luganda, Scots Gaelic, Tagalog, Turkish, Ukrainian, Yakut.